

# FFmpeg for Unity

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## 1. Overview

This assets is to execute [FFmpeg](#) on editor and application.

You can do below:

- Play Movies(mp4, avi, mov etc.)
- Capture Game View or Camera(In Game)
- Convert Video Files
- Live Streaming to [Youtube](#)(rtmp etc.)
- etc.

## 2. Correspondence Environment

- Unity Editor(Windows/Mac/Linux)
- Standalone(Windows/Mac/Linux)(Mono/IL2CPP)
- Android
- iOS

### 3. Note for FFmpeg

This assets uses FFmpeg binary(Windows/Mac/Linux) or plugins(Windows/Mac/Android/iOS) built below.

It using libraries is part only(please show Lisense). So, if libraries you want to use is not included, please rebuild and replace files.

For Windows and Mac (Library), replace generated 1. from 2. and execute it.

- Windows:
  1. <https://github.com/NON906/ffmpeg-windows-build-helpers>
  2. [https://github.com/NON906/fftools\\_lib](https://github.com/NON906/fftools_lib)
- Mac(Library):
  1. <https://github.com/NON906/ffmpeg-kit>
  2. <https://github.com/NON906/FfmpegUnityMacPlugin>
- Android/iOS: <https://github.com/NON906/ffmpeg-kit>
- Mac(Binary)/Linux: <https://github.com/NON906/ffmpeg-build-script>

### 4. How to Use

1. Import this assets.
2. (Optional) Rebuild FFmpeg and replace files.
3. Edit Settings:
  - 3-1. Open 'File -> Build Settings' on the menu bar.
  - 3-2. Click 'Player Settings...'.
    - 3-3. Change to '.NET 4.x' in 'Other Settings -> Api Compatibility Level'.
4. If want to run sample scenes, do it:
  - Please place `sample.mp4` to `Assets/StreamingAssets`.
  - PlayerVR, ListVR: In addition to the above, import the 'XR Interaction Toolkit' and its corresponding Plugin from Package Manager.
  - YoutubeLive: Please replace stream key(from Youtube Live) to `[STREAM_KEY]`.

NOTE: (For Android) From ver1.6, `com.athenica.smartexception` is included.

If you get an error after updating, please delete the setting of

`Assets/Plugins/Android/mainTemplate.gradle`.

## 5. Settings

You can set setting from Tools->Ffmpeg for Unity->Open Setting Window on the menu bar.  
If there are no malfunctions, I recommend using the default settings.

- Library (Windows/Mac)

Use ffmpeg library.

- Built In Binary (Windows/Mac/Linux)

Use the ffmpeg binary that exists in the following path.

When building, it will be installed automatically.

It will be the same operation as when Use Built In is turned ON in ver1.6.1 or earlier.

Windows: Assets/FfmpegUnity/Bin/Windows

Mac: Assets/FfmpegUnity/Bin/Mac

Linux: Assets/FfmpegUnity/Bin/Linux

- Installed Binary (Windows/Mac/Linux)

Use the ffmpeg binary installed to OS.

You need to have ffmpeg installed in advance and have a PATH.

It will be the same operation as when Use Built In is turned OFF in ver1.6.1 or earlier.

- Library Memory (Android/iOS)

Connect Unity and ffmpeg-kit with memory copy.

- Pipe (Android/iOS)

Connect Unity and ffmpeg-kit with named pipes.

It will be the same operation in ver1.6.1 or earlier.

NOTE:

- for Windows:  
If you want to use 'Built In Binary' or 'Installed Binary' on Windows, please import [HERE](#).
- for Mac:  
If you want to use 'Built In Binary' on Mac, please import [HERE](#).  
Please note that this binary may not work on M1 Macs, so if applicable, prepare a corresponding binary or use something other than 'Built In Binary'.

## 6. Scenes

- List
- ListVR

Displays a list of available functions.

The latter is for VR.

- Convert
- ConvertProgress

There are sample of video format conversion.

- Player
- PlayerVR

There are sample of video player.

- TexturePlayer
- TexturePlayerRender

There are sample of how to apply a video to Texture.

- Capture
- CaptureCamera
- CaptureRenderTexture

There are sample of video capture.

The capturing source is different for each.

- SendStream
- TextureStreamPlayer
- TexturePlayerCompare

There are sample of video streaming distribution.

- YoutubeLive

This is a sample of delivery to Youtube Live.

- BytesInput
- BytesOutput
- ConvertBytes
- TextureBytesPlayer
- TextureBytesCaptureAndPlayer

There are sample for inputting / outputting bytes directly.

- BatchTexture
- BatchTextureWithSound

This is a sample for processing frame by frame.

- GdigrabTexturePlayer

This is a sample using screen capture.

This is Windows only.

## 7. Component Details

### 7-1. Basic

#### 7-1-1. FfmpegCommand

Execute FFmpeg command.

Inspectors:

Execute On Start: Execute the command at the beginning of the scene. If not uses, please call `StartFfmpeg()` when want to start command.

Options: FFmpeg commands option. If you set below, this replace to path when execute command.

```
{STREAMING_ASSETS_PATH} : Application.streamingAssetsPath  
{PERSISTENT_DATA_PATH} : Application.persistentDataPath  
{TEMPORARY_CACHE_PATH} : Application.temporaryCachePath
```

NOTE: 'Use Built In' is moved to [Setting Window](#).

## 7-1-2. FfmpegPlayerCommand

Setting video to FfmpegPlayerVideoTexture, and audio to AudioSource.

Inspectors:

Execute On Start, Options: Same as FfmpegCommand.

Input Options: Ffmpeg commands option before input streams.

Default Path:

Default Path: Standard path. Both are the same:

```
Default Path: STREAMING_ASSETS_PATH  
Input Path: sample.mp4
```

```
Default Path: NONE  
Input Path: {STREAMING_ASSETS_PATH}/sample.mp4
```

Input Path: Video File path.

Auto Settings: Auto Setting to video size etc. In most cases ON.

Video Textures: Depending on the FfmpegPlayerVideoTexture settings, the following will be done:

```
FfmpegPlayerVideoTexture.VideoTexture is empty: Make Texture2D to VideoTexture.  
Please set with scripts to Renderer etc.
```

```
FfmpegPlayerVideoTexture.VideoTexture is setted in RenderTexture: Write to  
RenderTexture.
```

Audio Sources: Set audio to AudioSource's clip.

### **7-1-3. FfmpegCaptureCommand**

Capture video and audio, and save movies or streaming.

NOTE: When processing is not in time, video speed may be abnormally high. If that happens, please reduce the processing (change video size etc.).

Execute On Start, Options: Same as FfmpegCommand.

Capture Sources: Input for capture. If Video Size is 0 or less, this value is auto setting.

## 7-2. Bytes

### 7-2-1. FfmpegBytesCommand

It is bytes input/output version of FfmpegCommand.

Set the input bytes to `AddInputBytes(bytes, inputNo)`.

(`inputNo` is input stream's number corresponding to 'Input Option'.)

Get the output bytes from `GetOutputBytes(outputNo)`.

(Return value is `byte[]`)

(`outputNo` is output stream's number corresponding to 'Output Option'.)

Input Options: Number of input streams, and FFmpeg commands option before input streams.

NOTE: The number of this list matches the number of streams.

Even if you don't need the options, you need to prepare items for the number of streams (can be blank).

Output Options: Number of output streams, and FFmpeg commands option before output streams.

NOTE: Note the same as 'Input Options'.

### 7-2-2. FfmpegBytesPlayerCommand

It is bytes input version of FfmpegPlayerCommand.

Please check [FfmpegPlayerCommand](#) and [FfmpegBytesCommand](#).

### 7-2-3. FfmpegBytesCaptureCommand

It is bytes output version of FfmpegCaptureCommand.

Please check [FfmpegCaptureCommand](#) and [FfmpegBytesCommand](#).



## 7-3. Batch Per Frames

### 7-3-1. FfmpegGetTexturePerFrameCommand

This is the version of FfmpegPlayerCommand that can be obtained for each frame.

Call `GetNextFrame()` coroutine to update to the next frame.

(Does not advance to the next frame while not calling)

### 7-3-2. FfmpegWriteFromTexturesCommand

A component for writing the contents of Texture to a video.

Call `WriteTexture(inputTexture)` coroutine to write the contents of the specified Texture.